

Fig. 2: Actual-size PCB pattern of the low-cost quadraphonic amplifier

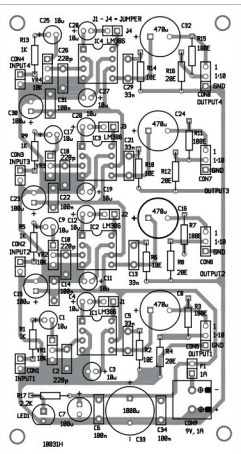


Fig. 3: Component layout of the PCB

of 4-ohm, 8-ohm and 16-ohm. But most parameters of the ICs are given only for a load of 8-ohm.

Generally speaking, if we accept the maximum power dissipation around 1.25W, we have the following approximate limitations:

1. With a load of 16-ohm, power supply could go up to 14V, or even 16V (more than 15V only for model LM386N-4)
2. With a load of 8-ohm to 12-ohm, up to 12V to 15V
3. With a load of 4-ohm, up to 9V, or even 12V
4. With a load of 2-ohm, up to 5V, or even 6V

Load of 2-ohm is not mentioned in the data sheet. So the risk of using it would be entirely upon by the user. LM386N-1 could work off +5V power supply from a USB port, if needed.

It is advised to use large cop-

per paths on the PCB to relieve the heat from the IC. Most frequently the IC is not loaded continuously to the maximum. But for musicians practicing heavy distortions, LM386 could be overloaded. Measures should be taken not to overheat the IC. Luckily, it is not easy to damage the IC by overloading.

Gain of LM386 could be set from 20 to 200. When jumpers J1, J2, J3 and J4 are open, gain of the corresponding amplifier is 20. If these jumpers are closed, gain of the corresponding IC is 200. LM386 has internal individual input resistors of around 50-kilo-ohm between inputs 2 and 3 and ground pin 4.

Each amplifier has two types of outputs. Main outputs are for loudspeakers. The loudspeakers should have impedance of at least 8-ohm and power rating of 2W, and

mounted in a wooden box. Outputs with resistive dividers (1:10) are for headphones. Connecting low-impedance headphones, for example, with 32-ohm or less, will not damage the outputs.

Construction and testing

An actual-size, single-side PCB pattern of the low-cost quadraphonic amplifier is shown in Fig. 2 and its component layout in Fig. 3.

The full power supply range of LM386 is available. Preferred power supply is stabilised with regulators such as 7805, 7806, 7808, 7809, 7812 and 7815, depending on the selected load resistances.

The circuit could work with low-cost wall adaptors that provide necessary voltage and at least 1A output aver-

age current. All filtering capacitors should not be omitted, even if there are such capacitors in the external power supply.

LM386 can work successfully off a 6V battery from a motorcycle or kids' electric car (typically 6.6V, maximum around 7.7V). The audio amplifier could work with one, two or three 4.5V batteries of type 3R12. Working with a single battery of 9V such as 9F22 is possible but not for a long time.

The amplifier gives output power up to around 1W per channel and has a wide range of applications. **EFY**



Petre Tzy Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFFPPT(Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria